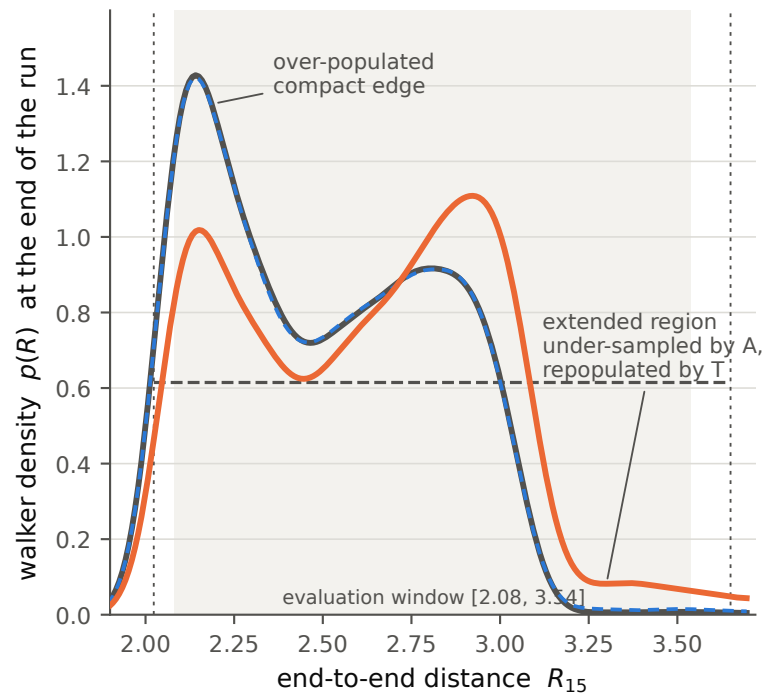


Pentane R_{15} , $\beta = 2$, 16 fresh seeds — gentle Wasserstein reallocation re-balances the marginal; the free-energy gain sits where sampling was added

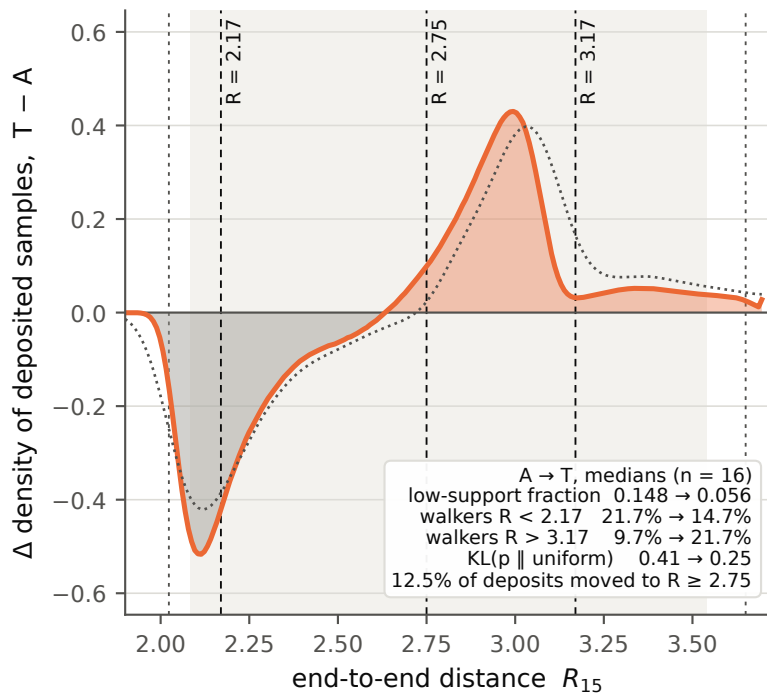
T vs A: $\Delta \hat{F}(C) = -16.0\%$ $[-16.4, -15.6]$, 16/16; D_{cond} unchanged (+0.3%). Torsional transitions and round trips unchanged; T contains no repair step — the mechanism is marginal sample reallocation.

Corrected mean-force reference v2; evaluation window $[2.08, 3.54]$ shaded; dotted verticals = OT target domain $[2.02, 3.65]$; the same frozen confirmatory raw outputs as the paired analysis, no reruns.

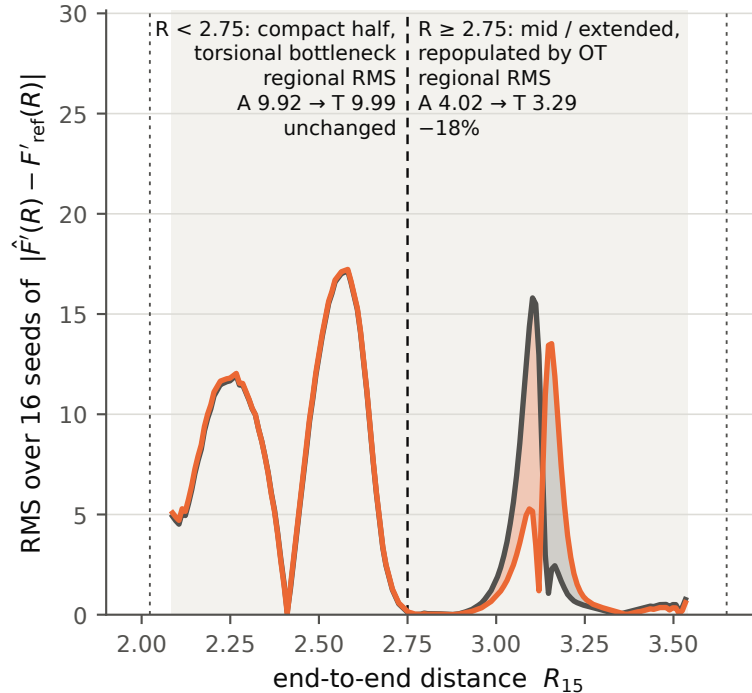
a Marginal allocation along R_{15}



b Where OT reallocates sampling effort



c Where free-energy information improves



--- uniform target on the OT domain - - - F uniform FR ($\approx$ A)
 — A ABF — T OT (capped, $\alpha = 0.01$)

more sampling under T deposits (raw accumulators, 16 seeds pooled)
 less sampling under T final walkers, $p_T - p_A$

T more accurate A more accurate — A ABF — T OT